recovered from the constituent branes. This insight, Strominger recently said, “has led some string theorists to want to claim victory—to claim that the information is recovered as black holes evaporate. In my opinion this conclusion is premature; there is still much work to be done in order to see if this is true.”11 Vafa concurs, saying that he “is agnostic on this question—it could still turn out either way.”12 Answering this question is a central goal of current research. As Hawking has put it,

Most physicists want to believe that information is not lost, as this would make the world safe and predictable. But I believe that if one takes Einstein’s general relativity seriously, one must allow for the possibility that spacetime ties itself in knots and that information gets lost in the folds. Determining whether or not information actually does get lost is one of the major questions in theoretical physics today.13

The second unresolved black hole mystery concerns the nature of spacetime at the central point of the hole.14 A straightforward application of general relativity, going all the way back to Schwarzschild in 1916, shows that the enormous mass and energy crushed together at the black hole’s center causes the fabric of spacetime to suffer a devastating rift, to be radically warped into a state of infinite curvature—to be punctured by a spacetime singularity. One conclusion that physicists drew from this is that since all of the matter that has crossed the event horizon is inexorably drawn to the center of the black hole, and since once there the matter has no future, time itself comes to an end at the heart of a black hole. Other physicists, who over the years have explored the properties of the black hole’s core using Einstein’s equations, revealed the wild possibility that it might be a gateway to another universe that tenuously attaches to ours only at a black hole’s center. Roughly speaking, where time in our universe comes to an end, time in the attached universe just begins.

We will take up some of the implications of this mind-boggling possibility in the next chapter, but for now we want to stress one important point. We must recall the central lesson: Extremes of huge mass and small size leading to unimaginably large density invalidate the sole use of Einstein’s classical theory and require that quantum mechanics be brought to bear as well. This leads us to ask, What does